Linear Algebra

•metrix•

Row ← صف

$$\begin{pmatrix} 1_{e_{11}} & 3_{e_{12}} \\ 2_{e_{21}} & 4_{e_{22}} \end{pmatrix}_{2\times 2} \quad m \times n$$

size = $(m \times n)$

Row, Coulm

Coulm
عمود

* Type of metrix

* Rectengular → $m \neq n$

Ahmed elsabbagh

$$A = \begin{pmatrix} 1 & 3 & 5 \\ 4 & 6 & 7 \end{pmatrix}_{2\times 3}$$

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}_{3\times 2}$$

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* Squre → $m = n$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}_{2\times 2} \qquad A = \begin{pmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{pmatrix}_{3\times 3}$$

* Diegonel → squre

$$A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7 \end{pmatrix}_{3\times3}$$

* Identity metrix (unit metrix)

→ squere metrix ell elements ere zeros except (Diegonel) = (ones)

→ I

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{2\times2}$$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}_{3\times3}$$

* I_2, I_3

Ex: Given $A \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ $B = \begin{pmatrix} 3 & x \\ 4 & y \end{pmatrix}$

Find value of x, y

if $A = B$

$\therefore x = 3$, $y = 4$

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* Vectors
  - colum $x = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}_{3\times1}$
  - Row $y = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}_{1\times3}$

* zero metrix (O_2)

$$O_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}_{2\times2}$$

* Addition / Subtraction

$$A_{m\times n} = B_{m\times n}$$

=

* scaler multiplication

$$5A = 5 \times \begin{pmatrix} x & x \\ x & x \end{pmatrix}$$

* multiplication

$$A_{m\times n} * B_{m\times n} = AB_{m\times n}$$

=

Ex: $A = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$

Find
* A+B
* A−B
* B−A
* AB

$$A+B = \begin{bmatrix} 7 & 6 & 5 \\ 0 & 1 & 3 \\ 7 & 3 & 7 \end{bmatrix}_{3\times 3}$$

$$A-B = \begin{bmatrix} -5 & 4 & -1 \\ 0 & -1 & -1 \\ -1 & 1 & 1 \end{bmatrix}_{3\times 3}$$

$$B-A = \begin{bmatrix} 5 & -4 & 1 \\ 0 & 1 & 1 \\ 1 & -1 & -1 \end{bmatrix}_{3\times 3}$$

- $A+B = B+A$
- $(A+B)+C = A+(B+C)$
- $K(A+B) = KA+KB$
- $A(BC) = (AB)C$
- $A(B+C) = AB+AC$
- $A+0 = 0+A = A$
- $AI = A$
- $A0 = 0$

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$AB =$

$$= \begin{bmatrix} 9 & 8 & 19 \\ -2 & 0 & 0 \\ 32 & 9 & 25 \end{bmatrix} \quad 3\times3$$

*Transpose of a metrix

$A \Longrightarrow A^T$

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \Longrightarrow A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$

$A = \begin{pmatrix} 3 & 5 \\ 7 & 9 \\ 0 & 10 \end{pmatrix} \Longrightarrow A^T = \begin{pmatrix} 3 & 7 & 0 \\ 5 & 9 & 10 \end{pmatrix}$

*Symmetric metrix

المصفوفة التي تساوى معكوسها

$C = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} \Longrightarrow C^T = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$

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*Determinants of metrix

$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = (1\times4) - (2\times3) = -2$

$+ \quad - \quad +$

$\begin{vmatrix} 2 & -3 & 4 \\ 1 & -2 & 1 \\ 3 & 1 & 2 \end{vmatrix} = +2\left[(-2\times2) - (1\times1)\right] - (3)(2-1)$

$+4(1+6) = 15$

#

*Inverse of a metrix

$A = A^{-1}$ $\quad AA^{-1} = A^{-1}A = I_n$

شروط ان ال metrix لها معكوس

$\det(A) \neq \text{zero}$

محدد المصفوفة

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 4 - 6 = -2 \neq \text{zero}$

Elementary row operations and Inverse of matrix

1- interchange two rows

$$\begin{bmatrix} 2 & 10 & 5 \\ 1 & 4 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & 3 \\ 2 & 10 & 5 \end{bmatrix}$$

2- multiply a row by a nonzero constant

$$\begin{bmatrix} 2 & 10 & 5 \\ 1 & 4 & 3 \end{bmatrix} \quad \text{Row } 1 \times \frac{1}{2}$$

$$\sim \begin{bmatrix} 1 & 5 & 2.5 \\ 1 & 4 & 3 \end{bmatrix}$$

3- Add a multiple of a row to another row

$$-2 \ -8 \ -6 \leftarrow \begin{bmatrix} 1 & 4 & 3 \\ 2 & 10 & 5 \end{bmatrix} \rightarrow \quad \text{Row } 1 \times (-2) + \text{Row } 2$$

$$\sim \begin{bmatrix} 1 & 4 & 3 \\ 0 & 2 & -1 \end{bmatrix}$$

$$A \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Ex: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$ Find A^{-1}

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$= \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$$

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* inverse of metrix 3×3

$$A^{-1} = \frac{1}{|A|} \, \text{edj}(A)$$

→ edjoint metrix (edj (A))

↳ is the transpose of the cofector metrix هى مدور مصفوفة العوامل المرافقة

$$\text{edj}(A) = (A^*)^T$$

EX: find the inverse of metrix $A = \begin{bmatrix} 1 & -4 & -8 \\ 2 & 1 & 5 \\ 3 & 2 & 4 \end{bmatrix}$

solution

$$|A| = \begin{vmatrix} \overset{+}{1} & \overset{-}{-4} & \overset{+}{-8} \\ 2 & 1 & 5 \\ 3 & 2 & 4 \end{vmatrix} = 1(4-10) - (-4)(8-15) + (-8)(4-3)$$

$$= -42 \neq \text{zero}$$

$$\text{Cof}(A) = A^* = \begin{bmatrix} +\begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} & -\begin{vmatrix} 2 & 5 \\ 3 & 4 \end{vmatrix} & +\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} \\ -\begin{vmatrix} -4 & -8 \\ 2 & 4 \end{vmatrix} & +\begin{vmatrix} 1 & -8 \\ 3 & 4 \end{vmatrix} & -\begin{vmatrix} 1 & -4 \\ 3 & 2 \end{vmatrix} \\ +\begin{vmatrix} -4 & -8 \\ 1 & 5 \end{vmatrix} & -\begin{vmatrix} 1 & -8 \\ 2 & 5 \end{vmatrix} & +\begin{vmatrix} 1 & -4 \\ 2 & 1 \end{vmatrix} \end{bmatrix}$$

$$A^* = \begin{bmatrix} +(-6) & -(-7) & +(1) \\ -(0) & +(28) & -(14) \\ +(-12) & -(21) & +(9) \end{bmatrix} = \begin{bmatrix} -6 & 7 & 1 \\ 0 & 28 & -14 \\ -12 & -21 & 9 \end{bmatrix}$$

$$\text{edj}(A) = (A^*)^T = \begin{bmatrix} -6 & 7 & 1 \\ 0 & 28 & -14 \\ -12 & -21 & 9 \end{bmatrix}^T \rightarrow \begin{bmatrix} -6 & 0 & -12 \\ 7 & 28 & -21 \\ 1 & -14 & 9 \end{bmatrix}$$

$$A^{-1} = \frac{-1}{42} \begin{bmatrix} -6 & 0 & -12 \\ 7 & 28 & -21 \\ 1 & -14 & 9 \end{bmatrix}$$

*invers matrix by using ERO.

*معكوس المصفوفة باستخدام عمليات الصف البسيطة

$$EX: A=\begin{bmatrix}4&0&0\\-2&3&0\\6&-1&5\end{bmatrix}$$

$$[A \,|\, I] \xrightarrow{\text{عمليات الصف}} [I \,|\, A^{-1}]$$

$$\left[\begin{array}{ccc|ccc}4&0&0&1&0&0\\-2&3&0&0&1&0\\6&-1&5&0&0&1\end{array}\right] \xrightarrow{\frac{1}{4}R_1}$$

$$\left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{4}&0&0\\-2&3&0&0&1&0\\6&-1&5&0&0&1\end{array}\right] \xrightarrow{2R_1+R_2}$$

$$\left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{4}&0&0\\0&3&0&\frac{1}{2}&1&0\\6&-1&5&0&0&1\end{array}\right] \xrightarrow{-6R_1+R_3}$$

$$\left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{4}&0&0\\0&3&0&\frac{1}{2}&1&0\\0&-1&5&-\frac{3}{2}&0&1\end{array}\right] \xrightarrow{\frac{1}{3}R_2} \left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{4}&0&0\\0&1&0&\frac{1}{6}&\frac{1}{3}&0\\0&-1&5&-\frac{3}{2}&0&1\end{array}\right]$$

$$\left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{4}&0&0\\0&1&0&\frac{1}{6}&\frac{1}{3}&0\\0&-1&5&-\frac{3}{2}&0&1\end{array}\right] \xrightarrow{R_2+R_3} \left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{4}&0&0\\0&1&0&\frac{1}{6}&\frac{1}{3}&0\\0&0&5&-\frac{4}{3}&\frac{1}{3}&1\end{array}\right]$$

$$\left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{4}&0&0\\0&1&0&\frac{1}{6}&\frac{1}{3}&0\\0&0&5&-\frac{4}{3}&\frac{1}{3}&1\end{array}\right] \xrightarrow{\frac{1}{5}R_3} \left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{4}&0&0\\0&1&0&\frac{1}{6}&\frac{1}{3}&0\\0&0&1&-\frac{4}{15}&\frac{1}{15}&\frac{1}{5}\end{array}\right]$$

$$\underbrace{\quad}_{I} \qquad \underbrace{\quad}_{A^{-1}}$$

Systems of Linear Equations

*Augmented matrix:

System

$$x - 4y + 3z = 5$$
$$-x + 3y - z = -3$$
$$2x - 4z = 6$$

$$A^* = [A \mid b]$$

Coefficient matrix

$$\begin{bmatrix} 1 & -4 & 3 \\ -1 & 3 & -1 \\ 2 & 0 & -4 \end{bmatrix}$$

Augmented matrix

$$\begin{bmatrix} 1 & -4 & 3 & 5 \\ -1 & 3 & -1 & -3 \\ 2 & 0 & -4 & 6 \end{bmatrix}$$

$$x_1 + 3x_2 - x_3 + x_4 = -7$$
$$x_1 + 2x_2 - 2x_4 = 9$$
$$3x_1 + x_2 + 4x_3 = 7$$

$$A = \begin{bmatrix} 1 & 3 & -1 & 1 \\ 1 & 2 & 0 & -2 \\ 3 & 1 & 4 & 0 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad B = \begin{bmatrix} -7 \\ 9 \\ 7 \end{bmatrix}$$

The Augmented is $[A|B]$

$$= \left[\begin{array}{cccc|c} 1 & 3 & -1 & 1 & -7 \\ 1 & 2 & 0 & -2 & 9 \\ 3 & 1 & 4 & 0 & 7 \end{array}\right]$$

*Method Gauss:

* Gauss Elimination and Gauss-Jordan Elimination

• Row-Echelon Form and Reduced Row Echelon form

*Row Echelon form

• أي صف كله أصفار موجود في نهاية المصفوفة.
(لو صفين أو ثلاثة لازم يكونوا في الآخر ورا بعض)

• أي صف يتكون من أصفار في أول عمود أو اثنين في الأكتر لازم يقابلنا بعدهم ① ـه يسمى Leading 1
لو رقم غير ① هضرب المصفوفة في رقم عشان يبقى ①

• لازم نلاقي Leading 1 ـه مرتبة من الشمال لليمين

*Reduced Row Echelon
له نفس خصائص Row Echelon Form لكن له خاصية زيادة:

• العمود اللي فيه Leading 1 يكون كله أصفار من فوق الـ ① ومن تحته

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Determine whether each matrix is in row Echelon form. If it is determine whether the matrix is also in reduced row Echelon form.

(a) leading 1

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

ref ✓
rref X

(b) Leading 1

$$\begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & -4 \end{bmatrix}$$

ref X
rref X

(c)

$$\begin{bmatrix} 1 & -5 & 2 & -1 & 3 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

ref ✓
rref X

(d)

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

موجود في الآخر

ref ✓
rref ✓

(e)

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

ref X
rref X

(f)

$$\begin{bmatrix} 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ref ✓
rref ✓

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{bmatrix}$$

ref X
rref X

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 2 & 1 \end{bmatrix}$$

ref X
rref X

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ref ✓
rref ✓

Liner system — Associated Augmented

$x - 2y + 3z = 9$

$-x + 3y = -4$

$2x - 5y + 5z = 17$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{array}\right]$$

نحولها

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

ref / Gauss-Elimination

نتعامل مع غالبا ← Gauss-jordan ←

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

rref

Ex: solve the system using Gauss jordan elimination

$-3x + 5y = -22$

$3x + 4y = 4$

$4x - 8y = 32$

solution

$$\left[\begin{array}{cc|c} -3 & 5 & 22 \\ 3 & 4 & 4 \\ 4 & -8 & 32 \end{array}\right] \xrightarrow{-\frac{1}{3}R_1} \begin{bmatrix} 1 & -\frac{5}{3} & -\frac{22}{3} \\ 3 & 4 & 4 \\ 4 & -8 & 32 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{5}{3} & -\frac{22}{3} \\ 3 & 4 & 4 \\ 4 & -8 & 32 \end{bmatrix} \xrightarrow{-3R_1+R_2 \text{ and } \frac{1}{4}R_3} \begin{bmatrix} 1 & -\frac{5}{3} & -\frac{22}{3} \\ 0 & 9 & -18 \\ 1 & -2 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{5}{3} & -\frac{22}{3} \\ 0 & 9 & -18 \\ 1 & -2 & 8 \end{bmatrix} \xrightarrow{-R_1+R_3} \begin{bmatrix} 1 & -\frac{5}{3} & -\frac{22}{3} \\ 0 & 9 & -18 \\ 0 & -\frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{5}{3} & -\frac{22}{3} \\ 0 & 9 & -18 \\ 0 & -\frac{1}{3} & \frac{2}{3} \end{bmatrix} \xrightarrow{\frac{1}{4}R_3 \text{ and } -3R_3} \begin{bmatrix} 1 & -\frac{5}{3} & -\frac{22}{3} \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -\frac{5}{3} & -\frac{22}{3} \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix} \xrightarrow{-R_2+R_3} \begin{bmatrix} 1 & -\frac{5}{3} & -\frac{22}{3} \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$Y = -2$ and $X = 4$

Rank of matrix $= 2$

∴ Leading 1 = variables

∴ Unique solution

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Metrix inversion.

مصفوفة مجهول

$A X = b \Longrightarrow X = A^{-1} b$

مصفوفة معاملات

مصفوفة ثوابت

Find x, y

$2x + y = 5$

$3x + 2y = 8$

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

1- $\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} = (2 \times 2) - (1 \times 3) = 4 \times 3 = 1$

2- $A^{-1} = \frac{1}{1} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

$X = A^{-1} b = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \times \begin{bmatrix} 5 \\ 8 \end{bmatrix}$

$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$\therefore x = 2$

$y = 1$

$x - 2y + z = -4$

$y + 2z = 4$

$2x + 3y + 2z = 2$

$$AX = b \Rightarrow \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \\ 2 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \\ 2 \end{bmatrix}$$

$X = A^{-1} b$

Find A^{-1} :

$$A^{-1} = \frac{1}{18} \begin{bmatrix} -8 & -1 & -5 \\ 4 & -4 & -2 \\ 2 & -7 & 1 \end{bmatrix}$$

$$X = A^{-1} b = \frac{1}{18} \begin{bmatrix} -8 & -1 & -5 \\ 4 & -4 & -2 \\ 2 & -7 & 1 \end{bmatrix} \times \begin{bmatrix} -4 \\ 4 \\ 2 \end{bmatrix}$$

$$= \frac{1}{-18} \begin{bmatrix} 18 \\ -36 \\ -18 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

* Note *

* عدد ال leeding 1 = عدد المتغيرات ← unique solution

$$\begin{pmatrix} - & - & - & - & - \\ - & - & - & - & - \\ 0 & 0 & 0 & 0 & = 2 \end{pmatrix}$$

رقم = صفرصفرصفر ←

No solution

* ال leeding 1 > عدد المتغيرات ← infinity solution

$(\lambda-5)(\lambda-10)=0 \quad \therefore \lambda_1=5 \quad , \quad \lambda_2=10$

Eigen vector:

* when $\lambda=5$

$$(A-\lambda I)X=0$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} X_{11} \\ X_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \left[\begin{array}{cc|c} 3 & 3 & 0 \\ 2 & 2 & 0 \end{array}\right] = \begin{bmatrix} X_{11} \\ X_{12} \end{bmatrix} \xrightarrow{\frac{1}{3}R_1}$$

$$\begin{array}{ccc} X_{11} & X_{12} & \\ \end{array}$$
$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 2 & 2 & 0 \end{array}\right] = \begin{bmatrix} X_{11} \\ X_{12} \end{bmatrix}$$

$$X_{11}+X_{12}=0$$

$$X_{11}=-X_{12}$$

Let $X_{11}=1 \quad \therefore X_{12}=-1$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ or } \begin{bmatrix} 2 \\ -2 \end{bmatrix} \text{ or } \begin{bmatrix} 3 \\ -3 \end{bmatrix}$$

* Deigonelize metrix:

$A = PDP^{-1}$

Eigen vector [illegible]

Eigenvelue

$P^{-1} \rightarrow \frac{1}{\det(P)} * adj(P)$

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

Note: $A^K = \begin{bmatrix} \lambda_1^K & 0 & 0 \\ 0 & \lambda_2^K & 0 \\ 0 & 0 & \lambda_3^K \end{bmatrix}$ $\quad A^K = PD^KP^{-1}$

Ex: if $A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}$ find Deigonelize A and find A^3

Solution

$\lambda_1 = 5 \quad \lambda_2 = 10 \Rightarrow$ Eigenvelue

$V_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 2 \end{bmatrix} \Rightarrow$ Eigen Vector

V_2

$P = [V_1 \; V_2] \Rightarrow \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}$

$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \Rightarrow \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}$

* Deigonelize $A = PDP^{-1} = \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix} * \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix} * \frac{1}{5} * \begin{bmatrix} 2 & -3 \\ 1 & 1 \end{bmatrix}$

\# A^3: $A^3 = PD^3P^{-1} = \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix} * \left(\begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}\right)^3 * \frac{1}{5} \begin{bmatrix} 2 & -3 \\ 1 & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix} * \begin{bmatrix} 5^3 & 0 \\ 0 & 10^3 \end{bmatrix} * \frac{1}{5} \begin{bmatrix} 2 & -3 \\ 1 & 1 \end{bmatrix} =$

$= \begin{bmatrix} 650 & 525 \\ 350 & 415 \end{bmatrix}$ #

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* Ceyley-Hemilton : نستبدل λ ب A

Ex: Verify ceyley-Hemilton theorem for given metrix $A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}$

And hence find a formule to A^{-1}

Solution

$$\det(A - \lambda I) = 0$$

$$\det\left(\begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = 0$$

$$\det\begin{pmatrix} 8-\lambda & 3 \\ 2 & 7-\lambda \end{pmatrix}$$

$$(8-\lambda) * (7-\lambda) - 2 * 3 = 0$$

$$\rightarrow 56\lambda^2 - 15\lambda - 50 = 0 \qquad \lambda \rightarrow A$$

$$56A^2 - 15A - 50 = 0$$

$$\begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}^2 - 15\begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix} + 50I = 0$$

$$= \begin{bmatrix} 70 & 45 \\ 30 & 105 \end{bmatrix} - \begin{bmatrix} 120 & 45 \\ 30 & 105 \end{bmatrix} + \begin{bmatrix} 50 & 0 \\ 0 & 50 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

Find formule A^{-1}

$A^2 - 15A + 50I = 0$ $\quad (\times A^{-1})$ each seid

$A - 15I + 50A^{-1} \Rightarrow 50A^{-1} = 15I - A$

$$\therefore A^{-1} = \frac{1}{50}(15I - A)$$

Ly=b

$$\begin{bmatrix}1&0&0\\2&1&0\\2&2&1\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}=\begin{bmatrix}6\\18\\18\end{bmatrix}=\begin{bmatrix}6\\6\\-6\end{bmatrix}$$

$\rightarrow y_1=6$

$\rightarrow 2y_1+y_2=18 \quad \therefore 2\times6+y_2=18 \quad \therefore y_2=6$

$\rightarrow 2y_1+2y_2+y_3=18 \quad \therefore 2\times6+2\times6+y_3=18 \quad \therefore y_3=-6$

$$\begin{bmatrix}2&1&2\\0&2&1\\0&0&3\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}6\\6\\-6\end{bmatrix}=\begin{bmatrix}3\\4\\-2\end{bmatrix}$$

$3x_3=-6 \rightarrow \therefore x_3=-2$

$2x_2+x_3=6 \rightarrow 2x_2+(-2)=6 \quad \therefore x_2=4$

$2x_1+x_2+2x_3=6 \rightarrow 2x_1+4+(-4)=6$

$\therefore x_3=3$

$|A|=|L|\cdot|U|=\prod_{i=1}^{n}\text{diag}\,L\cdot\prod_{i=1}^{n}\text{diag}\,U$

$=[1\times1\times1]\cdot[2\times2\times3]=12$ #

Factorization (LU Decomposition)

If $A=\begin{bmatrix}2&1&2\\4&4&5\\4&6&9\end{bmatrix}$ when $Ax=\begin{bmatrix}6\\18\\18\end{bmatrix}$

Decompose the matrix A into LU Decomposition

Find Eigenvalues of L

Eigen~~vector~~values of U

|A|

Solution

$$L=\overset{L}{\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}}\overset{U}{\begin{bmatrix}2&1&2\\\boxed{4}&4&5\\\boxed{4}&\boxed{6}&9\end{bmatrix}}\xrightarrow{R_2=-2R_1+R_2}$$

$$=\begin{bmatrix}1&0&0\\2&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}2&1&2\\0&2&1\\4&6&9\end{bmatrix}\xrightarrow{R_3=-2R_1+R_3}$$

$$=\begin{bmatrix}1&0&0\\2&1&0\\2&0&1\end{bmatrix}\begin{bmatrix}2&1&2\\0&2&1\\0&4&5\end{bmatrix}\xrightarrow{R_3=-2R_2+R_3}$$

$$\begin{bmatrix}1&0&0\\2&1&0\\2&2&\end{bmatrix}\begin{bmatrix}2&1&2\\0&2&1\\&&\end{bmatrix}$$

Vector spece

$\oplus \to$ جمع

$\odot \to$ ضرب

① Addition

$\to \alpha \oplus \beta = \beta + \alpha$

$\to \lambda \oplus (\alpha \oplus \beta) = (\lambda \oplus \beta) + \alpha =$

$= \beta \oplus (\lambda + \alpha)$

$\to \alpha \oplus (-\alpha) = \ominus \to$ zero vector

$\to \alpha \oplus \ominus = \alpha$

② seceler miltiple ction

$\to \alpha \odot 1 = \alpha$

$\to (s \odot t) \odot \alpha = s \odot (t \odot \alpha)$

$\to t \odot (\alpha \oplus \beta) = (t \odot \alpha) \oplus (t \odot \beta)$

$\to (s \oplus t) \odot \alpha = (s \odot \alpha) \oplus (t \odot \alpha)$

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Vector subspece

$W \subset\subset U$ [W is a subspece of U] if and only if

① $U + V \in W$ whenever $U, V \in W$

② $\alpha \cdot U \in W$ whenever $\alpha \in F$

Ex: If $A = \{(x, y, 0)\} \in R^3 \mid x, y \in R\}$ is $A \subset\subset R^3$?

$U = (x_1, y_1, 0)$ $V = (x_2, y_2, 0)$

① $U + V = [x_1 + x_2, y_1 + y_2, 0] \in A$

② $\alpha \cdot U = \alpha[x_1, y_1, 0] = [\alpha x, \alpha y_2, 0] \in A$

$\therefore A \subset\subset R^3$

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$S=\{(x,1,z)\}\in R^3 \,/\, x,y\in R\}$ Is $S\subset R^3$?

$U=(x_1,1,z_1)$ $\quad V=(x_2,1,z_2)$

$U+V=[x_1+x_2,\ 2,\ z_1+z_2]$

$\therefore A \not\subset \not\subset R^3$

*Linear combination

$\to W=U+V$

$U=\alpha_1 V_1+\alpha_2 V_2+\cdots \alpha_n V_n$

$V=x_1 V_1+x_2 V_2+\cdots+x_n V_n$

$U=(1,2,-1)$ $\quad V=(6,4,2)$

is R^3 show that

① $w=(9,2,7)$ Linear combination of

V and U.

$w=-3U+2U$

$=[-3,-6,3]+[12,8,4]=[9,2,7]$